

Affine rigidity of vector quasi-arithmetic means on closed convex domains with nonempty interior

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Abstract

Let $C \subset \mathbb{R}^n$ be closed and convex with nonempty Euclidean interior, and let $f : C \rightarrow \mathbb{R}^n$ be strictly monotone. We prove that the vector quasi-arithmetic mean defined by its extended monotone left inverse agrees with the arithmetic mean for one fixed positive weight vector if and only if $f(x) = Ax + b$ on all of C , where the symmetric part of A is positive definite. No continuity, measurability, differentiability, or convexity of $f(C)$ is assumed. This extends Tóth's whole-space classification and answers Open Problem 1 in arXiv:2608.19997v1. The proof combines his interior-fiber lemma with a local bound from monotonicity, an explicit fixed-weight Jensen argument, and exact interpolation to the boundary.

1 Statement and context

Throughout, $\langle \cdot, \cdot \rangle$ and $\|\cdot\|$ denote the Euclidean inner product and norm, and $B(p, r)$ is the open ball with center p and radius r . A map $f : C \rightarrow \mathbb{R}^n$ is *strictly monotone* if

$$\langle f(x) - f(y), x - y \rangle > 0 \quad (x, y \in C, x \neq y). \quad (1.1)$$

Write $D = \text{conv}(f(C))$, where conv is the ordinary finite convex hull, without an additional closure. For a nonempty closed convex C and a map satisfying (1.1), Tóth's inverse theorem [3, Theorem 2] supplies a unique map $T : D \rightarrow C$ with

$$T(f(x)) = x \quad (x \in C), \quad (1.2)$$

$$\langle T(u) - T(v), u - v \rangle \geq 0 \quad (u, v \in D). \quad (1.3)$$

This is the extended monotone left inverse, also denoted by $f^{(-1)}$. For $m \geq 2$ and fixed weights $\lambda_i > 0$ with $\sum_{i=1}^m \lambda_i = 1$, the associated mean is

$$\mathcal{M}_{f,\lambda}(x_1, \dots, x_m) = T \left(\sum_{i=1}^m \lambda_i f(x_i) \right).$$

Convexity of D makes this expression well defined.

Theorem 1. *Let $n \geq 1$, let $C \subset \mathbb{R}^n$ be closed and convex with $C^\circ \neq \emptyset$, and let $f : C \rightarrow \mathbb{R}^n$ satisfy (1.1). Fix an integer $m \geq 2$ and positive real weights $\lambda_1, \dots, \lambda_m$ summing to one, and let T be the extended monotone left inverse above. The following are equivalent:*

1. *For every $(x_1, \dots, x_m) \in C^m$,*

$$T\left(\sum_{i=1}^m \lambda_i f(x_i)\right) = \sum_{i=1}^m \lambda_i x_i. \quad (1.4)$$

2. *There are $A \in \mathbb{R}^{n \times n}$ and $b \in \mathbb{R}^n$ such that*

$$f(x) = Ax + b \quad (x \in C), \quad \langle Ah, h \rangle > 0 \quad (0 \neq h \in \mathbb{R}^n). \quad (1.5)$$

The matrix condition is positive definiteness of $(A + A^\top)/2$; it does not require symmetry of A . Only the single weight vector and arity in the statement are assumed. The conclusion includes every boundary point of C and applies to bounded and unbounded domains.

Tóth proves the equivalence for $C = \mathbb{R}^n$ [3, Theorem 4] and asks whether nonempty interior suffices [3, Open Problem 1]. Theorem 1 gives that domain extension. The singleton-fiber argument below is his Proposition 4, reproduced with attribution. The regularity arguments are elementary ingredients, not separate originality claims. Earlier vector means with continuous generators and convex image appear in Leonetti [2]; his Theorem 2.1 concerns density of iterated means. Weak affinity and its regularity are also studied by Boros and Tóth [1]. We give the required fixed-weight regularity proof in full. Zeng's work on convex-lens spectral constants [4] supplies broader operator-theoretic context; no spectral-set result enters the argument here.

2 Interior fibers and local regularity

Assume (1.4), and set $U = C^\circ$, $t = \lambda_1$ and $s = 1 - t$. Then $0 < t, s < 1$. Using the tuple $(x, y, \dots, y)$ with the original weights gives

$$T(tf(x) + sf(y)) = tx + sy \quad (x, y \in C). \quad (2.1)$$

Lemma 2 (Tóth [3, Proposition 4]). *Under (1.4), for every $p \in U$ one has*

$$\{u \in D : T(u) = p\} = \{f(p)\}.$$

Proof. Suppose $T(u) = p$ and $u \neq f(p)$. Put $c = \|u - f(p)\| > 0$ and $h = (u - f(p))/c$. Choose $a_0 > 0$ such that $p + ah \in C$ for $0 < a < a_0$. Comparing $f(p + ah)$ and u in (1.3) gives

$$0 \leq \langle ah, f(p + ah) - u \rangle = a(\langle f(p + ah) - f(p), h \rangle - c).$$

Thus $q(a) := \langle f(p + ah) - f(p), h \rangle \geq c$ on this interval. Its infimum ν is positive and finite: it is bounded below by c and above by any one of the finite values $q(a)$. Since $s < 1$, choose $\beta \in (0, a_0)$ with $q(\beta) < \nu/s$.

Let $z = p + \beta h$, $w = p + (s\beta/2)h$ and $v = tf(p) + sf(z) \in D$. By (2.1) and (1.2), $T(v) = p + s\beta h$ and $T(f(w)) = w$. Therefore

$$\begin{aligned} 0 &\leq \langle T(v) - T(f(w)), v - f(w) \rangle \\ &= \frac{s\beta}{2} (sq(\beta) - q(s\beta/2)) < 0, \end{aligned}$$

because $0 < s\beta/2 < a_0$ and $q(s\beta/2) \geq \nu$. This contradiction proves the assertion, since (1.2) already gives $T(f(p)) = p$. $\square$

In particular, (2.1) and Lemma 2 imply

$$f(tx + sy) = tf(x) + sf(y) \quad \text{if } x, y \in C \text{ and } tx + sy \in U. \quad (2.2)$$

Indeed, both sides belong to the fiber over $tx + sy$.

Lemma 3. *Let $p \in U$ and choose $r > 0$ with $\overline{B}(p, r) \subset U$. For the standard coordinate vectors e_j , set*

$$M_0 = \max_{\substack{1 \leq j \leq n \\ \varepsilon \in \{-1, 1\}}} \|f(p + \varepsilon r e_j)\|.$$

Then

$$\|f(p + u)\| \leq (2\sqrt{n} + 1)M_0 \quad \left(\|u\| \leq \frac{r}{2\sqrt{n}} \right). \quad (2.3)$$

Proof. Put $x = p + u$ and $v = f(x)$. If $v = 0$, the bound is immediate. Otherwise choose a coordinate j and sign ε with $\varepsilon v_j \geq \|v\|/\sqrt{n}$, and set $y = p + \varepsilon r e_j$. The weak consequence of (1.1) and the Cauchy–Schwarz inequality give

$$\left(\frac{r}{\sqrt{n}} - \|u\| \right) \|v\| \leq \langle v, y - x \rangle \leq \langle f(y), y - x \rangle \leq M_0(r + \|u\|).$$

The left coefficient is at least $r/(2\sqrt{n}) > 0$. Division and the assumed bound on $\|u\|$ yield (2.3). $\square$

The finite bound above uses no regularity of f . The following lemma explains explicitly how boundedness and one fixed-weight identity imply affinity.

Lemma 4. *Let $V \subset \mathbb{R}^n$ be nonempty, open and convex, and let $F : V \rightarrow \mathbb{R}^d$. Fix $0 < t < 1$ and $s = 1 - t$. Suppose that*

$$F(tx + sy) = tF(x) + sF(y) \quad (x, y \in V). \quad (2.4)$$

If F is bounded on a neighborhood of one point of V , then F is the restriction of a real affine map.

Proof. Choose $p \in V$, $R > 0$ and $K \geq 0$ such that $B(p, R) \subset V$ and

$$g(h) := F(p + h) - F(p), \quad \|g(h)\| \leq K \quad (\|h\| < R).$$

Taking one argument of (2.4) equal to p gives

$$g(th) = tg(h), \quad g(sh) = sg(h) \quad (\|h\| < R). \quad (2.5)$$

Set $\delta = \min(t, s)R/2$, so $0 < \delta \leq R/4$. If $\|u\|, \|v\| < \delta$, then $\|u/t\|, \|v/s\| < R/2$ and $\|u + v\| < 2\delta \leq R/2$. Applying (2.4) to $p + u/t$ and $p + v/s$, followed by (2.5), is consequently legitimate and yields

$$g(u + v) = tg(u/t) + sg(v/s) = g(u) + g(v). \quad (2.6)$$

For a positive integer N and $\|h\| < \delta/N$, all partial sums jh , $1 \leq j \leq N$, have norm less than δ . Induction using (2.6) gives $g(Nh) = Ng(h)$, and hence

$$\|g(h)\| \leq K/N \quad (\|h\| < \delta/N). \quad (2.7)$$

For any $\epsilon > 0$, choose $N > K/\epsilon$. Equation (2.7) proves continuity at zero.

For $v \in \mathbb{R}^n$ define

$$H(v) = Ng(v/N), \quad N \in \mathbb{N}, \quad \|v\|/N < \delta. \quad (2.8)$$

This is independent of the chosen positive integer N . Indeed, if N and L are both admissible, repeated addition of $v/(NL)$, with all partial sums inside $B(0, \delta)$, gives

$$g(v/N) = Lg(v/(NL)), \quad g(v/L) = Ng(v/(NL)).$$

Multiplying by N and L , respectively, proves independence in (2.8). Given $a, b \in \mathbb{R}^n$, choose one N large enough for a/N , b/N and $(a+b)/N$ to lie in $B(0, \delta)$. Equations (2.6) and (2.8) then imply $H(a+b) = H(a) + H(b)$. Since $H = g$ on $B(0, \delta)$, H is continuous at zero, and additivity makes it continuous everywhere. Additivity also gives $H(qv) = qH(v)$ for rational q . Approximating an arbitrary real scalar by rationals shows that H is real linear. Thus $H(v) = Av$ for a real d -by- n matrix A , and

$$F(p+h) = F(p) + Ah \quad (\|h\| < \delta). \quad (2.9)$$

To propagate this formula, fix $x \in V$ and put $x_k = p + t^k(x-p)$ for $k \geq 0$. Convexity ensures $x_k \in V$. Equation (2.4), applied to x_k and p , gives inductively

$$F(x_k) - F(p) = t^k(F(x) - F(p)).$$

For sufficiently large k , $t^k\|x-p\| < \delta$. Substituting (2.9) on the left and dividing by $t^k > 0$ yields $F(x) = F(p) + A(x-p)$ for every $x \in V$. $\square$

3 Boundary interpolation and completion

Proof of Theorem 1. First assume (1.4). The interior U is convex: if $x, y \in U$ and $0 < a < 1$, a ball $B(x, r) \subset C$ gives

$$B(ax + (1-a)y, ar) = aB(x, r) + (1-a)y \subset C.$$

Thus (2.2) holds for every $x, y \in U$. Lemma 3 supplies local boundedness, and Lemma 4, with $d = n$, yields $A \in \mathbb{R}^{n \times n}$ and $b \in \mathbb{R}^n$ such that

$$f(w) = Aw + b \quad (w \in U). \quad (3.1)$$

Now let $x \in C$ be arbitrary, and fix $y \in U$ with $B(y, \epsilon) \subset C$. The point $z = tx + sy$ is interior because

$$B(z, s\epsilon) = tx + sB(y, \epsilon) \subset C.$$

We may therefore use (2.2) even if x is on the boundary. Both z and y satisfy (3.1), so

$$\begin{aligned} tf(x) &= f(z) - sf(y) \\ &= A(tx + sy) + b - s(Ay + b) = t(Ax + b). \end{aligned}$$

Division by $t > 0$ proves the affine formula on all of C , without a boundary limit.

Fix $p \in U$ and $h \neq 0$. For some $a > 0$, $p + ah \in U$. Strict monotonicity now gives

$$0 < \langle f(p+ah) - f(p), ah \rangle = a^2 \langle Ah, h \rangle,$$

which proves the matrix condition in (1.5).

Conversely, assume (1.5). Then A is invertible, since $Ah = 0$ for $h \neq 0$ would violate the quadratic-form condition. The map f is strictly monotone, and $f(C) = AC + b$ is convex. Hence $D = f(C)$, and (1.2) determines $T(u) = A^{-1}(u - b)$ for every $u \in D$. This inverse is monotone: writing $u - v = Ah$ gives

$$\langle T(u) - T(v), u - v \rangle = \langle h, Ah \rangle \geq 0.$$

Finally,

$$T\left(\sum_{i=1}^m \lambda_i f(x_i)\right) = A^{-1}\left(A \sum_{i=1}^m \lambda_i x_i\right) = \sum_{i=1}^m \lambda_i x_i,$$

as required. $\square$

4 Scope and limitations

Nonempty *Euclidean* interior is essential. For the segment $C = \{(a, 0) : 0 \leq a \leq 1\} \subset \mathbb{R}^2$, take $f(a, 0) = (a, a^2)$, a special case of Tóth's example [3, Example 2]. Its strict-monotonicity pairing is $(a - b)^2$, while on $D = \text{conv}(f(C))$ the map $T(u, v) = (u, 0)$ is a monotone left inverse: the pairing for two inputs is $(u - u')^2$. It returns every weighted arithmetic barycenter, although f is not affine. Thus relative interior cannot replace the hypothesis as stated.

The positive-definite symmetric part need not exhaust A : for example,

$$A = \begin{pmatrix} 1 & -2 \\ 2 & 1 \end{pmatrix} \quad \text{satisfies} \quad \langle Ah, h \rangle = \|h\|^2.$$

Theorem 1 concerns equality with the arithmetic mean. It makes no classification claim for two arbitrary generators, for infinite-dimensional domains, or for degenerate weights.

The contribution extends the classification in [3] to the stated domains, including all boundary values. A targeted primary-literature screen on September 7, 2026 found no equivalent completed extension among the inspected statements; this does not establish exhaustive priority. Only the publisher record and abstract of [1] were inspected. That reference supplies context and no unproved dependency.

Computational and AI assistance. OpenAI Codex assisted with proof development, hypothesis checks, literature screening, and writing. A separate model context reconstructed the analytic argument. This is neither external peer review nor a proof-assistant certificate. The theorem rests on the displayed deductive proof, with no numerical search, statistical fit, or finite computation used to establish it.

References

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